

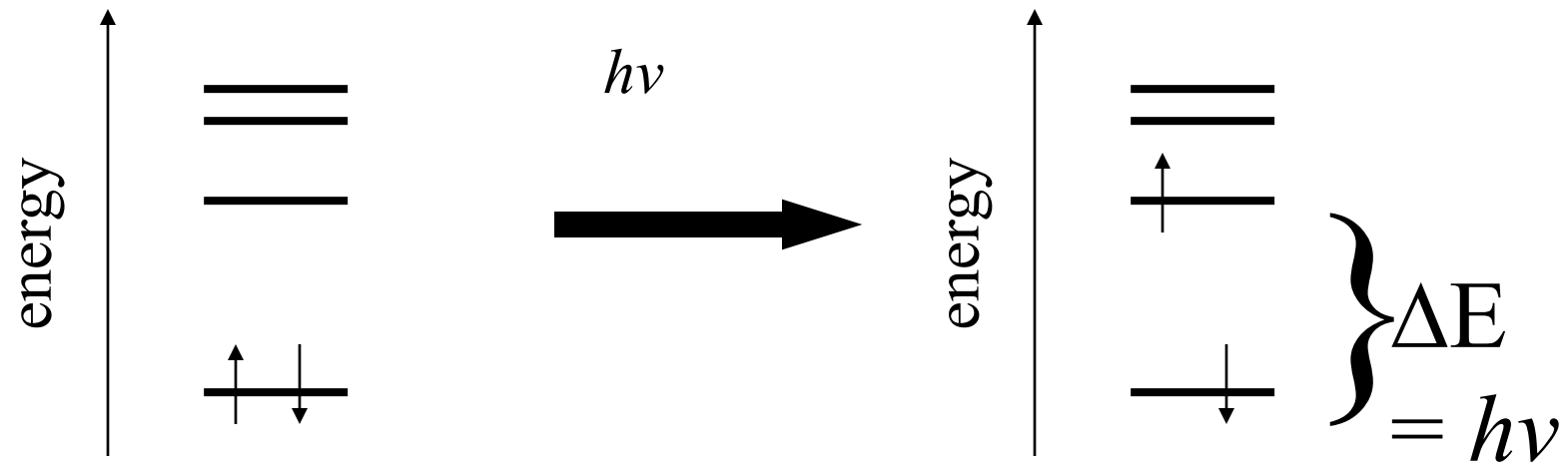
UV-Vis Spectroscopy

- UV- organic molecules
 - Outer electron bonding transitions
 - conjugation
- Visible – metal/ligands in solution
 - d-orbital transitions
- Instrumentation

Characteristics of UV-Vis spectra of Organic Molecules

- Absorb mostly in UV unless highly conjugated
- Spectra are broad, usually too broad for qualitative identification purposes
- Excellent for quantitative Beer's Law-type analyses
- The most common detector for an HPLC

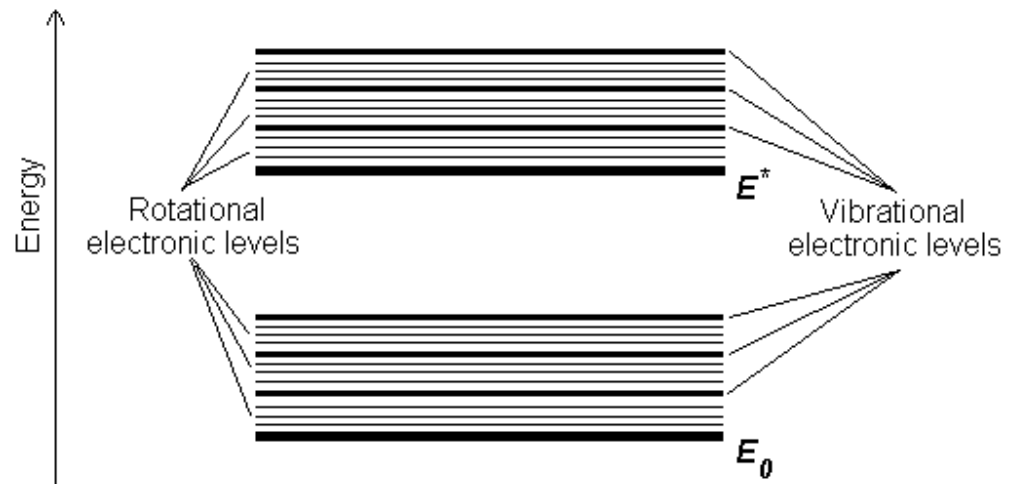
Molecules have quantized energy levels:
ex. electronic energy levels.

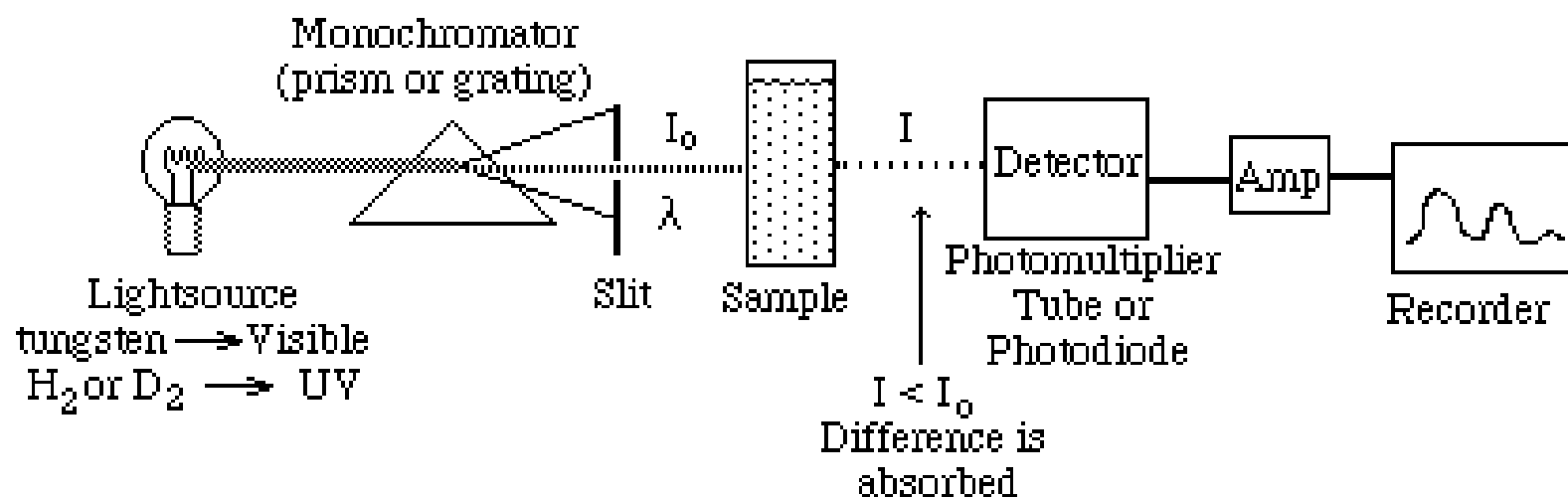


Q: Where do these quantized energy levels come from?

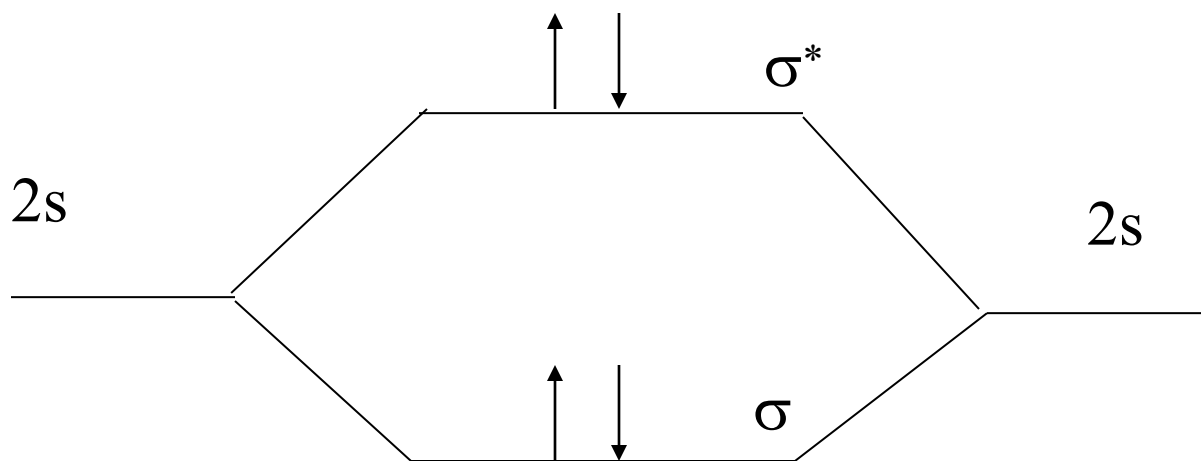
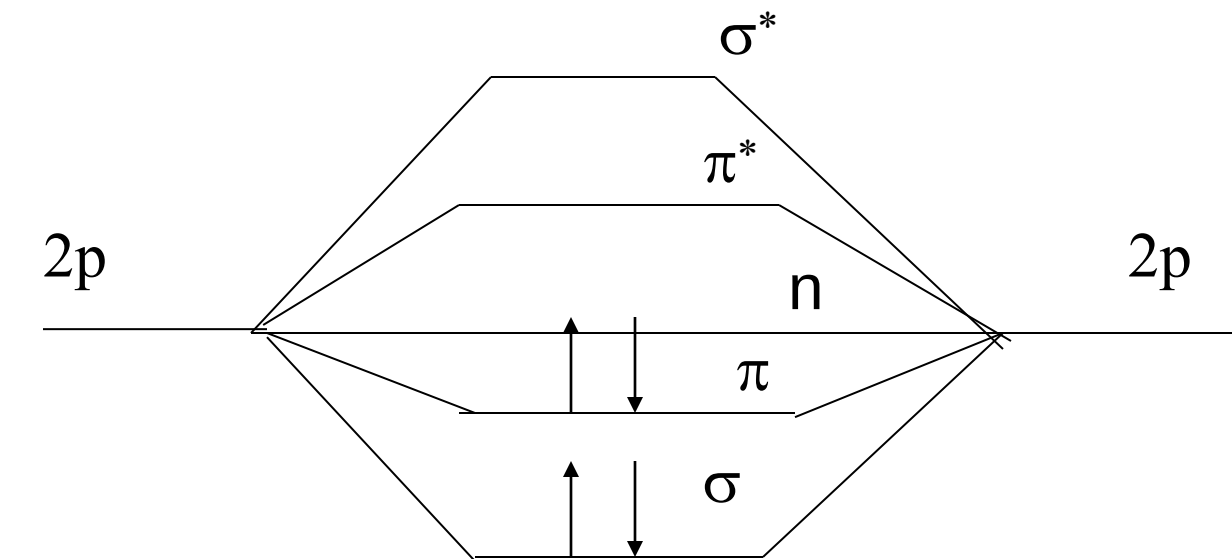
A: The electronic configurations associated with bonding.

Each electronic energy level (configuration) has associated with it the many vibrational energy levels we examined with IR.

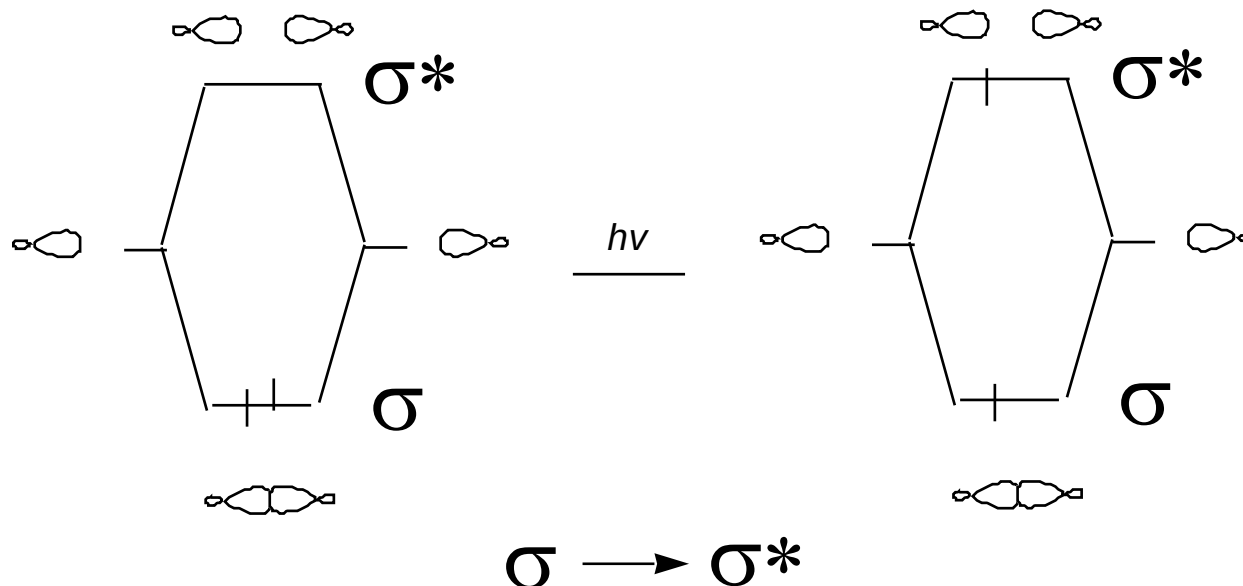
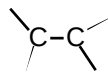




Molecular Orbital Theory



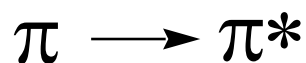
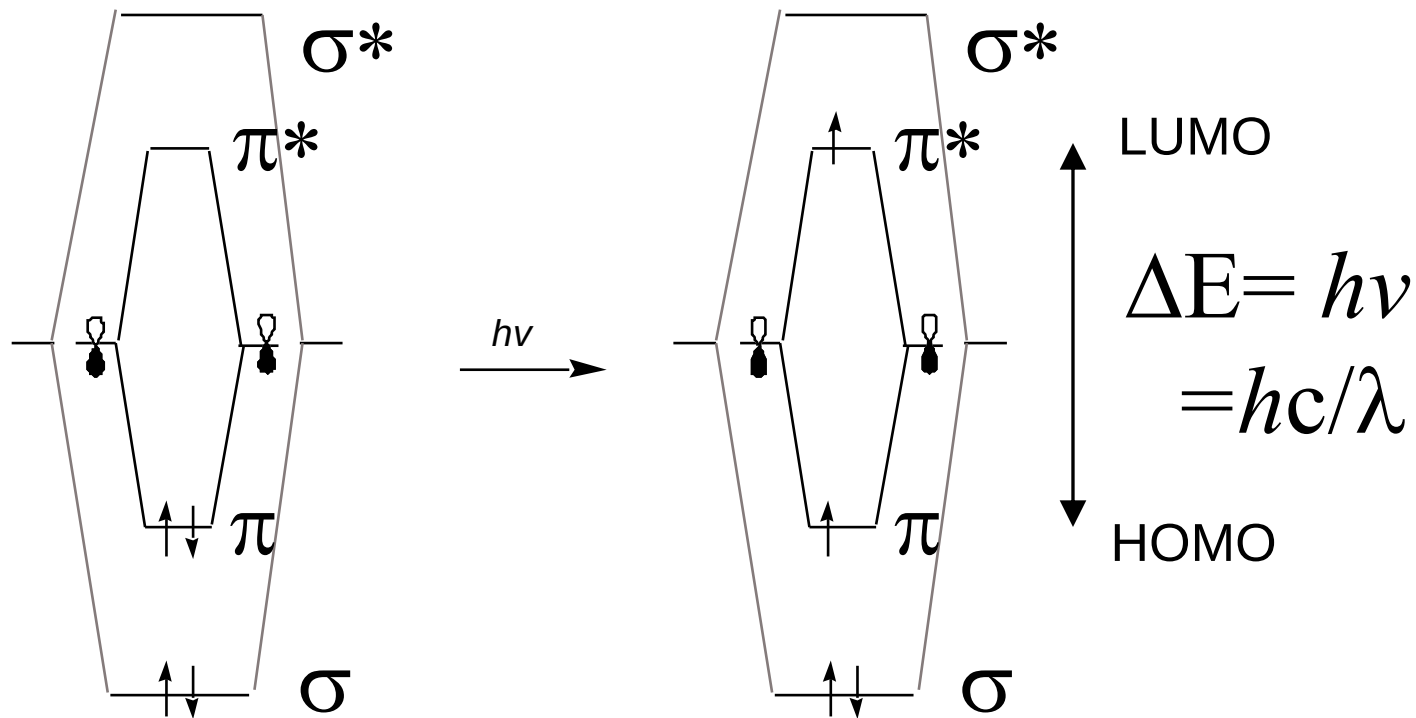
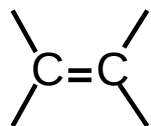
Ethane



$$\lambda_{\max} = 135 \text{ nm} \quad (\text{a high energy transition})$$

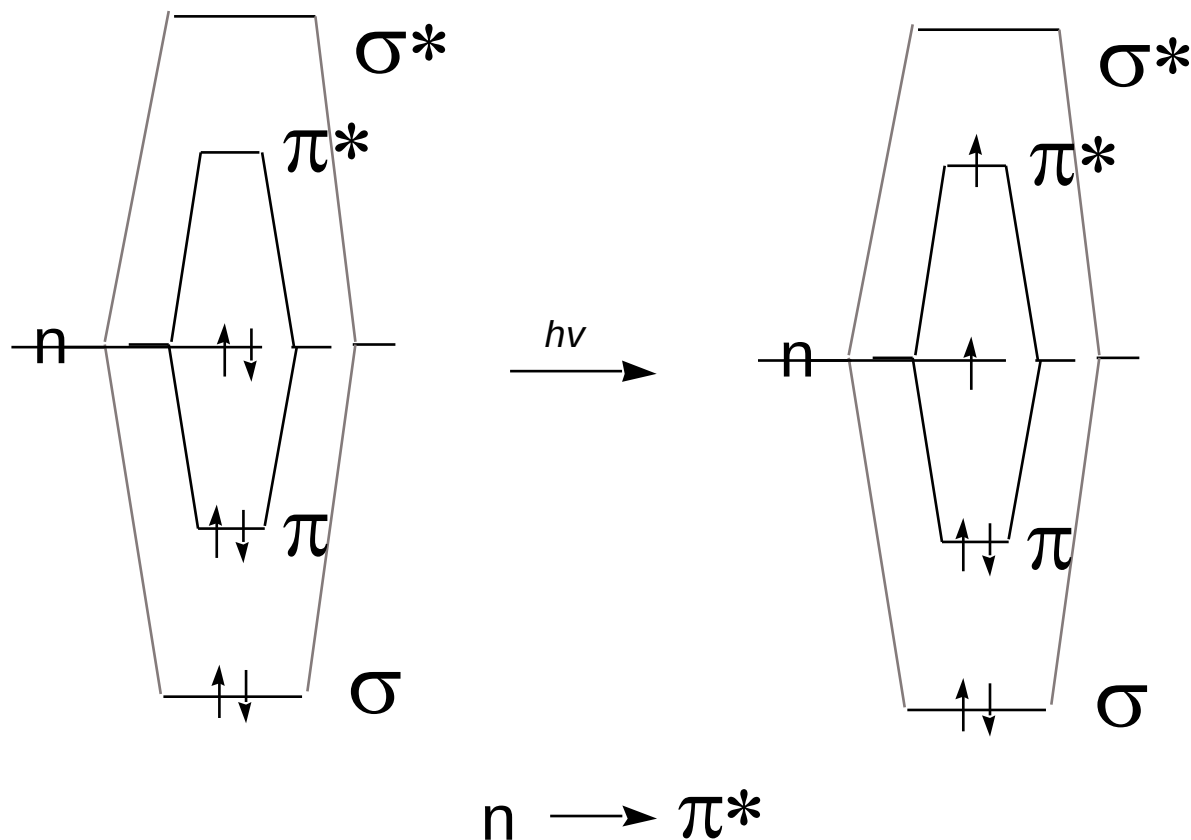
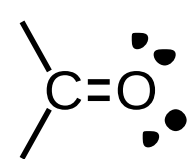
Absorptions having $\lambda_{\max} < 200 \text{ nm}$ are difficult to observe because everything (including quartz glass and air) absorbs in this spectral region.

Preferred transition is between
Highest Occupied Molecular Orbital (HOMO) and Lowest Unoccupied
Molecular Orbital (LUMO).



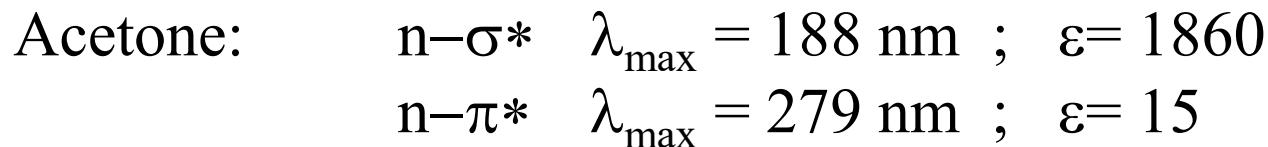
Example: ethylene absorbs at longer wavelengths:

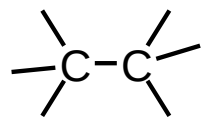
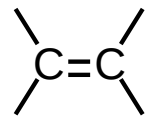
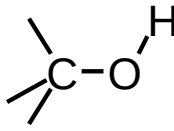
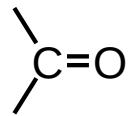
$$\lambda_{\max} = 165 \text{ nm } \epsilon = 10,000$$

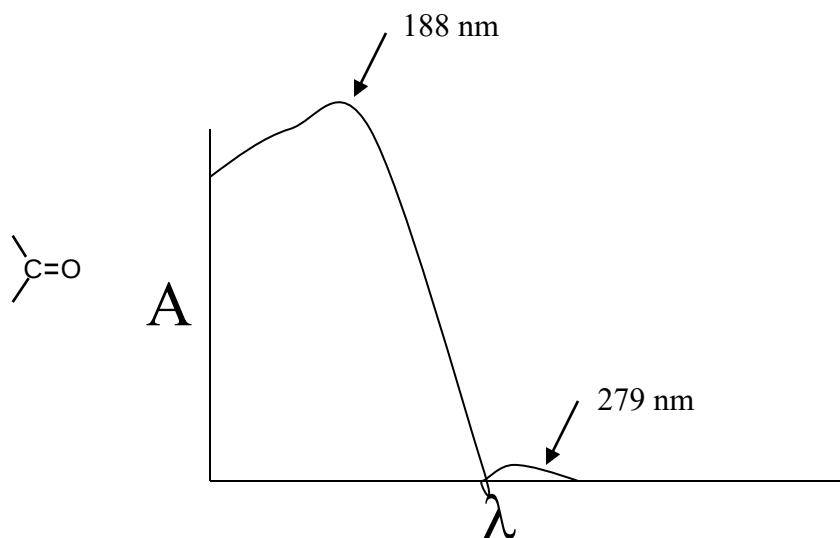


The n to π^* transition is at even lower wavelengths but is not as strong as π to π^* transitions. It is said to be “forbidden.”

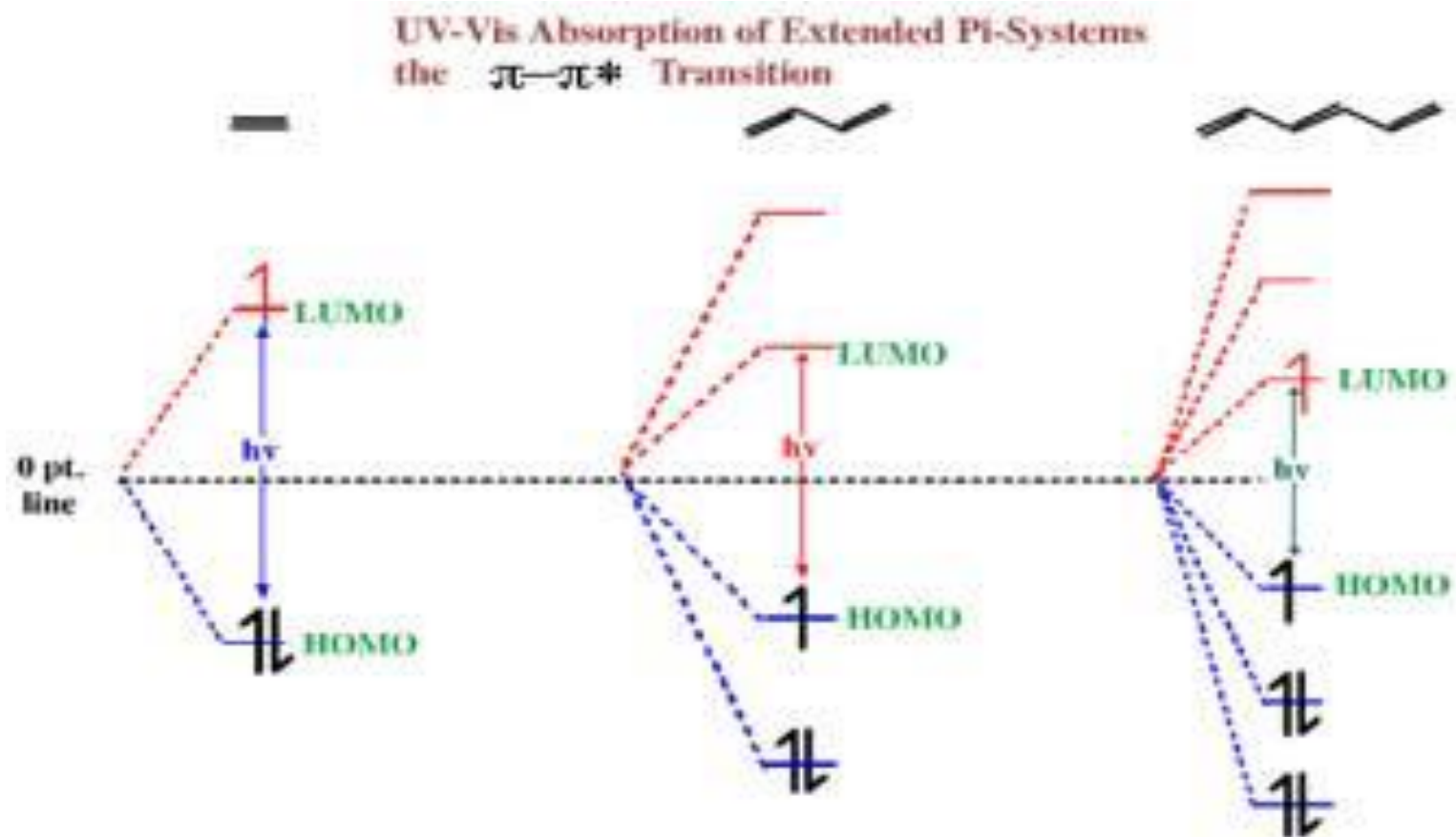
Example:



	$\sigma \rightarrow \sigma^*$	135 nm	
	$\pi \rightarrow \pi^*$	165 nm	
	$n \rightarrow \sigma^*$	183 nm	weak
	$\pi \rightarrow \pi^*$ $n \rightarrow \sigma^*$ $n \rightarrow \pi^*$	150 nm 188 nm 279 nm	weak



Conjugated systems:



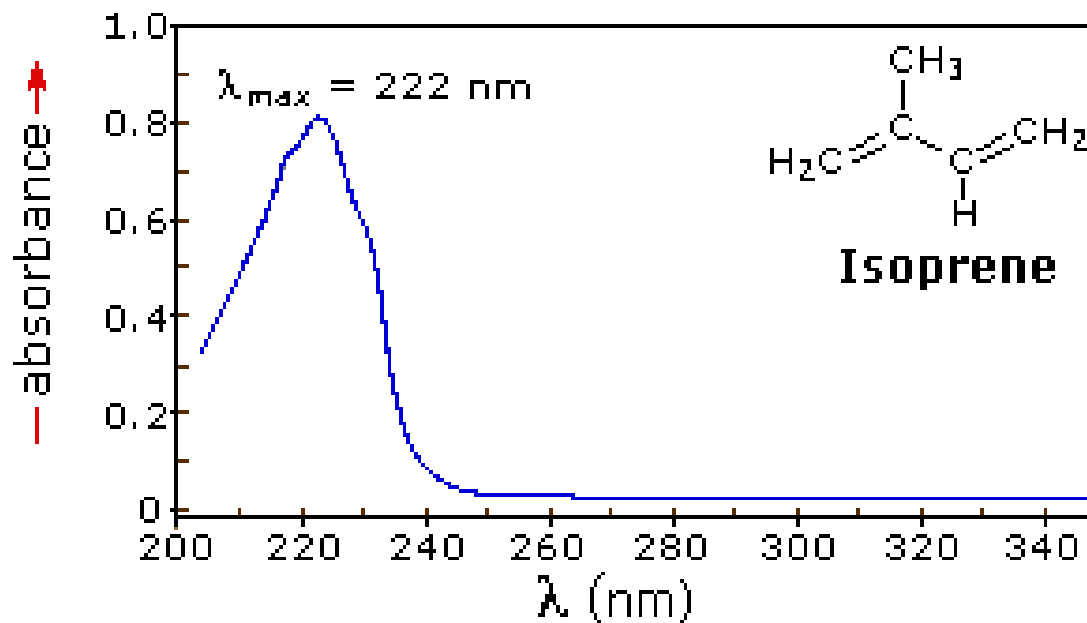
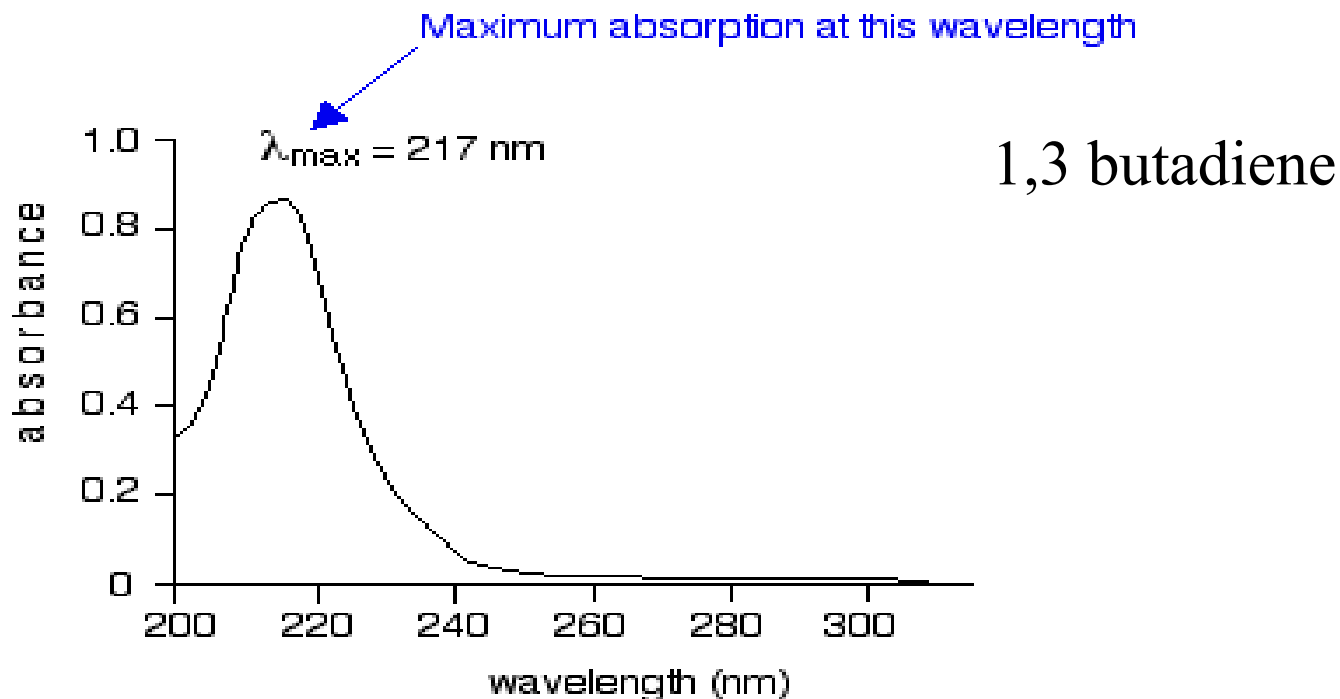
Preferred transition is between HOMO and LUMO

Note: Additional conjugation (double bonds) lowers the HOMO-LUMO energy gap:

Example:

1,3 butadiene: $\lambda_{\text{max}} = 217 \text{ nm}$; $\epsilon = 21,000$

1,3,5-hexatriene $\lambda_{\text{max}} = 258 \text{ nm}$; $\epsilon = 35,000$



Typical Absorptions of Simple Isolated Chromophores

Class	Transition	$\lambda_{\max}$ (nm)	$\log \epsilon$	Class	Transition	$\lambda_{\max}$ (nm)	$\log \epsilon$
R—OH	$n \rightarrow \sigma^*$	180	2.5	R—NO ₂	$n \rightarrow \pi^*$	271	<1.0
R—O—R	$n \rightarrow \sigma^*$	180	3.5	R—CHO	$\pi \rightarrow \pi^*$	190	2.0
R—NH ₂	$n \rightarrow \sigma^*$	190	3.5		$n \rightarrow \pi^*$	290	1.0
R—SH	$n \rightarrow \sigma^*$	210	3.0	R ₂ CO	$\pi \rightarrow \pi^*$	180	3.0
R ₂ C=CR ₂	$\pi \rightarrow \pi^*$	175	3.0		$n \rightarrow \pi^*$	280	1.5
R—C≡C—R	$\pi \rightarrow \pi^*$	170	3.0	RCOOH	$n \rightarrow \pi^*$	205	1.5
R—C≡N	$n \rightarrow \pi^*$	160	<1.0	RCOOR'	$n \rightarrow \pi^*$	205	1.5
R—N=N—R	$n \rightarrow \pi^*$	340	<1.0	RCONH ₂	$n \rightarrow \pi^*$	210	1.5

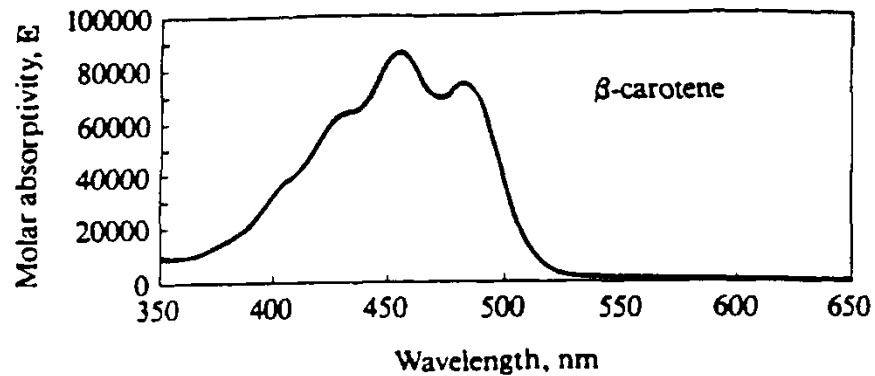
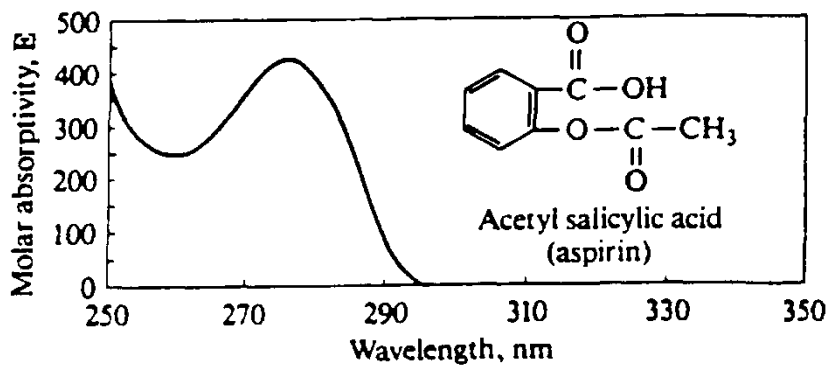
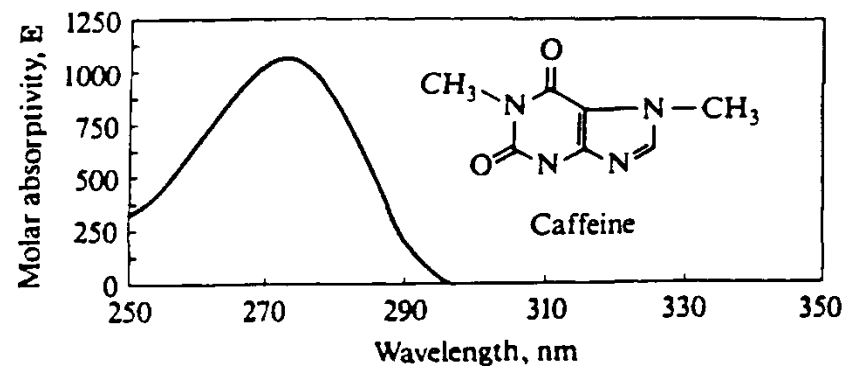
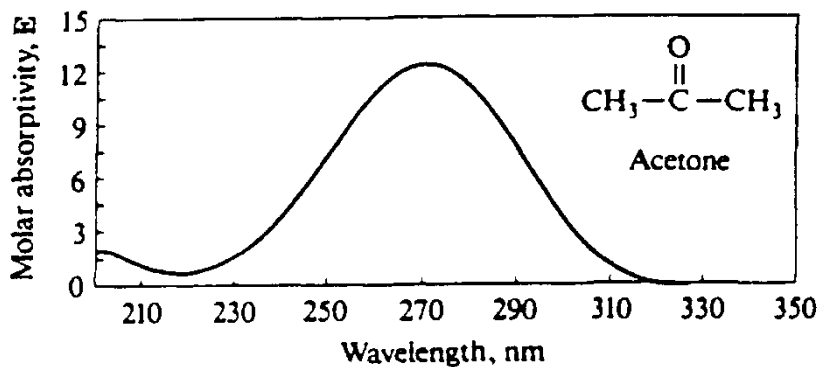
Absorption Characteristics of Some Common Organic Chromophores

Chromophore	Example	Solvent	$\lambda_{\max}$, nm	$\epsilon_{\max}$
Alkene	$\text{C}_6\text{H}_{13}\text{CH}=\text{CH}_2$	<i>n</i> -Heptane	177	13,000
Conjugate alkene	$\text{CH}_2=\text{CHCH}=\text{CH}_2$	<i>n</i> -Heptane	217	21,000
Alkyne	$\text{C}_5\text{H}_{11}\text{C}\equiv\text{C}-\text{CH}_3$	<i>n</i> -Heptane	178	10,000
			196	2,000
			225	160
Carbonyl	$\begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{CCH}_3 \end{array}$	<i>n</i> -Hexane	186	1,000
			280	16
	$\begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{CH} \end{array}$	<i>n</i> -Hexane	180	Large
			293	12
Carboxyl	$\begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{CNH}_2 \\ \text{CH}_3\text{COOH} \end{array}$	Water	214	60
		Ethanol	204	41
Azo	$\text{CH}_3\text{N}=\text{NCH}_3$	Ethanol	339	5
Nitro	CH_3NO_2	Isooctane	280	22
Nitrate	$\text{C}_2\text{H}_5\text{ONO}_2$	Dioxane	270	12
Nitroso	$\text{C}_4\text{H}_9\text{NO}$	Ethyl ether	300	100
			665	20
Aromatic	Benzene	<i>n</i> -Hexane	204	7,900
			256	200

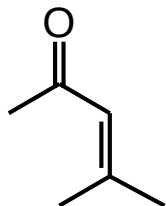
UV / visible Spectroscopy

- It is often difficult to extract a great deal of information from a UV spectrum by itself.
- Generally you can only pick out conjugated systems.
- Always use in conjunction with nmr and infrared spectra.
- As structural changes occur in a chromophore it is difficult to predict exact energy and intensity changes.
- Use empirical rules.: Woodward-Fieser Rules for dienes, Woodward's Rules for enones

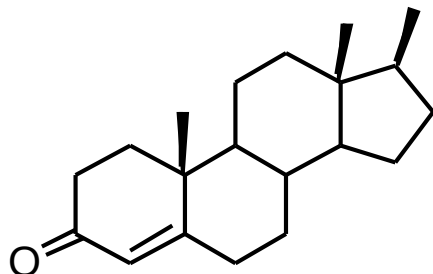
UV / visible Spectroscopy



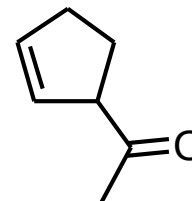
Similar structures have similar UV spectra:



$$\lambda_{\max} = 238, 305 \text{ nm}$$

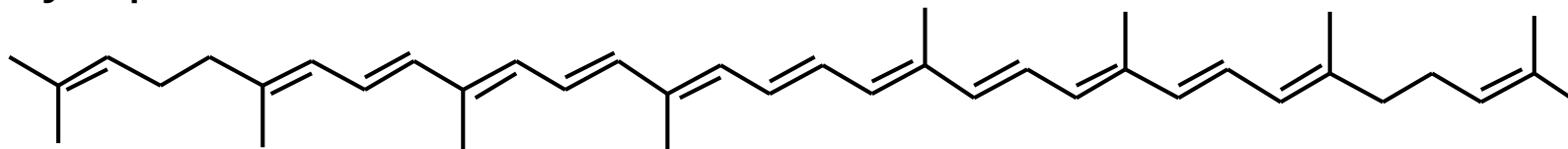


$$\lambda_{\max} = 240, 311 \text{ nm}$$



$$\lambda_{\max} = 173, 192 \text{ nm}$$

Lycopene:



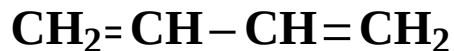
$$\lambda_{\max} = 114 + 5(8) + 11*(48.0 - 1.7*11) = 476 \text{ nm}$$

$$\lambda_{\max}(\text{Actual}) = 474.$$

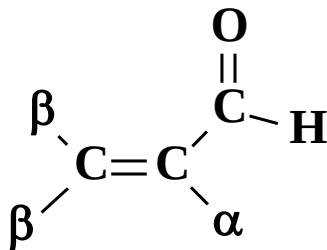
Woodward-Fieser Rules

for Predicting $\lambda_{\max}$

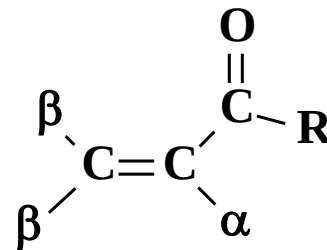
- The $\lambda_{\max}$ of the $\pi \rightarrow \pi^*$ transition for compounds with ≤ 4 conjugated double bonds can be calculated using **Woodward-Fieser rules**.



$$\lambda_{\max} = 217 \text{ nm}$$



$$\lambda_{\max} = 210 \text{ nm}$$



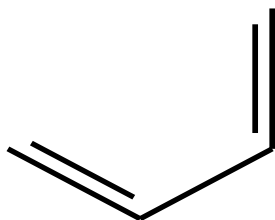
$$\lambda_{\max} = 215 \text{ nm}$$

General base value = 215

Base value:



transoid diene 214 nm

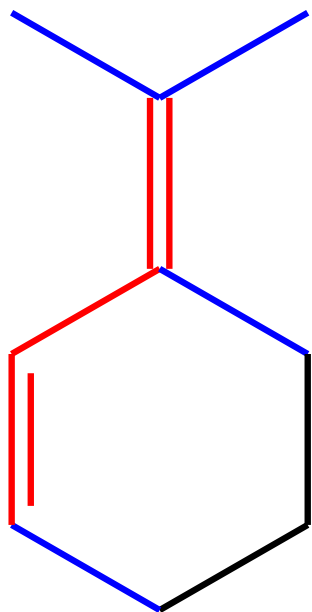


cisoid diene 253 nm

214 + 39 Homoannular system

Double bond extending conjugation	30 nm
Exocyclic double bond	5 nm
Alkyl group	5 nm
Cl, Br	5 nm
OH, OR	6 nm
SH, SR	30 nm
NH ₂ , NHR, NR ₂	60 nm

Determination of λ_{max}



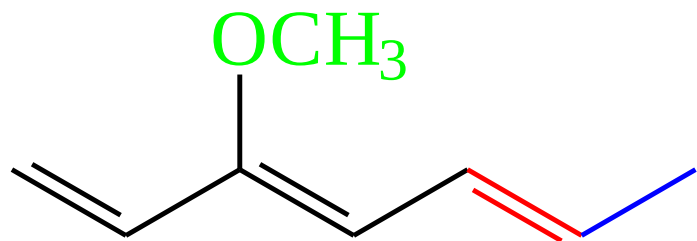
Base: 214 nm

R groups: 4 x 5 = 20 nm

exocyclic db = 5 nm

predicted λ_{max} = **239 nm**

Determination of λ_{max}



Base: 214 nm

double bond ext. conj. = 30 nm

OR group = 6 nm

R group = 5 nm

predicted λ_{max} = 255 nm

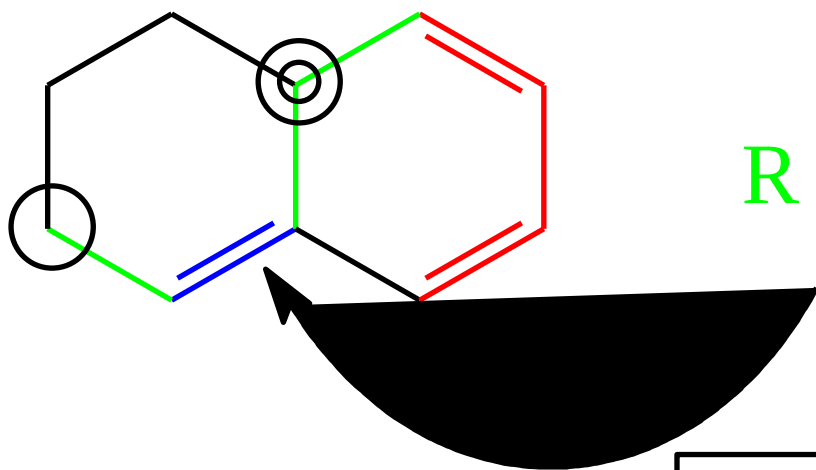
Determination of $\lambda_{\max}$

Base: 253 nm

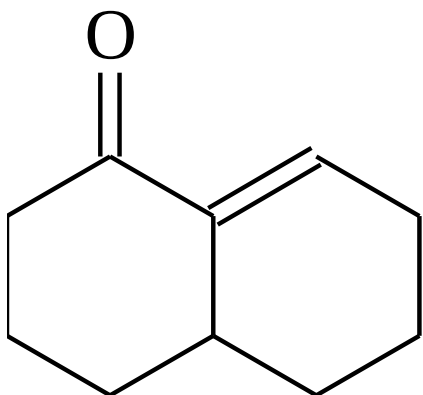
db ext. conj. = 30

R groups: 3 x 5 = 15

exocyclic db = 5



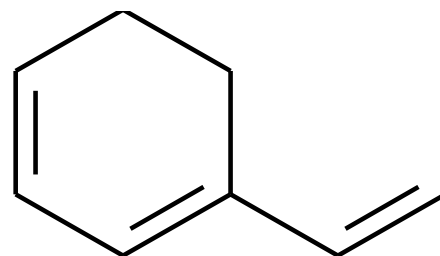
predicted $\lambda_{\max}$ = **303 nm**



Base	= 215
α substituent	= 10
β substituent	= 12
Exocyclic db bd	= 5

Calculated	= 242
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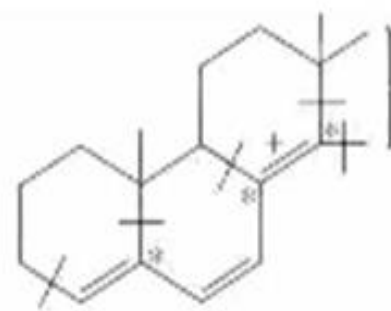
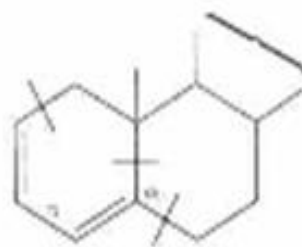
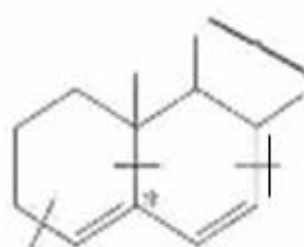
Observed	= 241
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Base	= 253
Additional db bd	= 30
2 alkyl substituents	= 10

Calculated	= 293
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Observed	= 293
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Parent

214

214

214

alkyl substituent (5)

$$\times 3 = 15$$

$$\times 3 = 15$$

$$\times 5 = 25$$

exocyclic (5)

$$\times 1 = 5$$

$$\times 1 = 5$$

$$\times 3 = 15$$

extra conjugation (30)

$$\times 0$$

$$\times 0$$

$$\times 1 = 30$$

homoannular (39)

$$\times 0$$

$$\times 1 = 39$$

$$\times 0$$

Predicted $\lambda_{\max}$ (nm)

234

273

284

Observed* $\lambda_{\max}$ (nm)

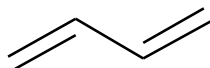
235

275

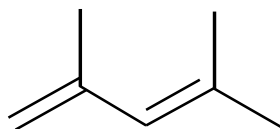
283

Conjugation lowers the energy differences between the HOMO and LUMO energy levels, and so conjugated dienes absorb at *longer wavelengths* than isolated dienes, and trienes absorb at *longer wavelengths* than dienes, etc.

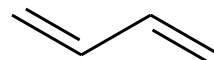
Alkyl group substitution on double bonds also causes absorption to occur at longer wavelengths.



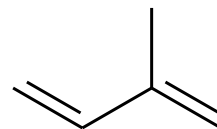
$$\lambda_{\text{max}} = 217\text{nm}$$



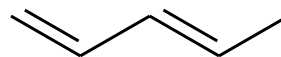
$$\lambda_{\text{max}} = 232\text{nm}$$



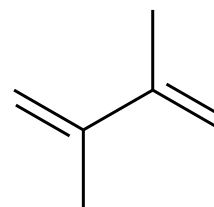
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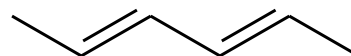
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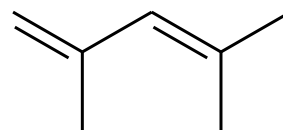
223



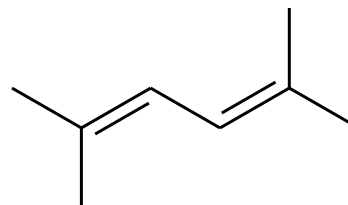
226



227



232



240

Quantitative Spectroscopy

- Beer Lambert Law

$$A = \log \left(\frac{I_0}{I} \right) \epsilon \ell c$$

A = Absorbance

I_0 = intensity (power per unit area) of the incident radiation

I = intensity (power per unit area) of the transmitted radiation

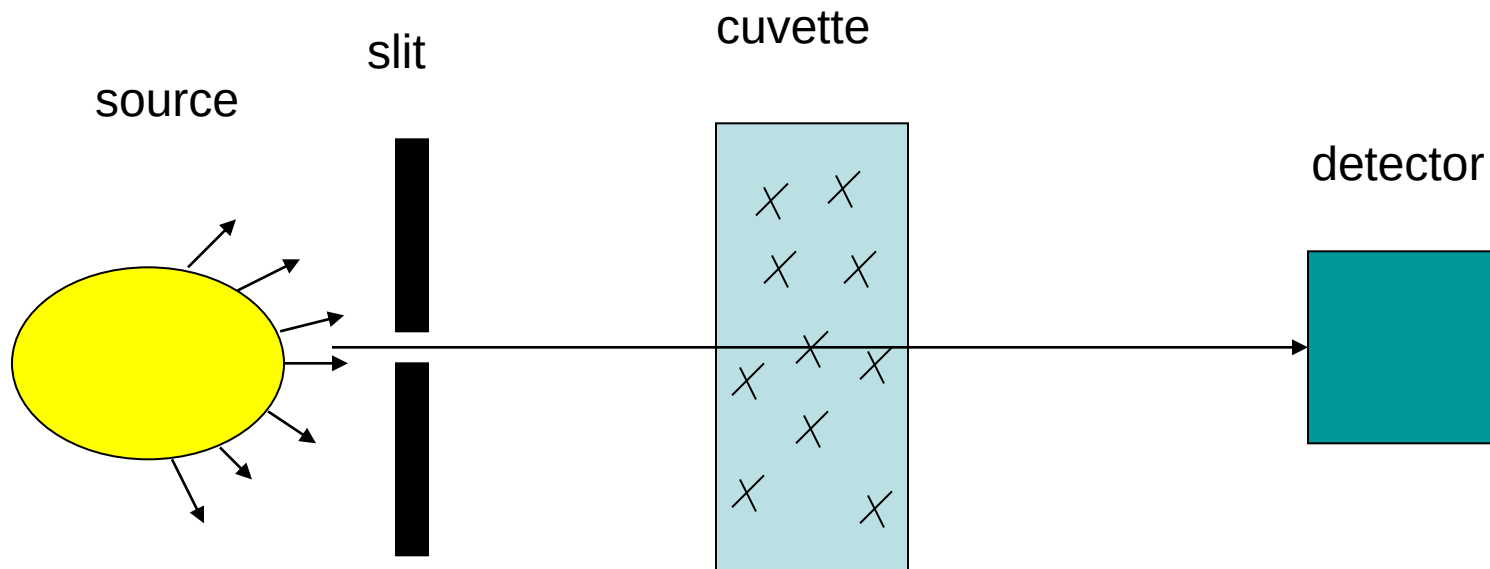
ϵ = molar absorptivity (unique for a given compound)

ℓ = path length

c = concentration

$$A = -\log T$$

T = Transmittance



- $A = -\log T = \log(P_0/P) = ebc$
- $T = P_{\text{solution}}/P_{\text{solvent}} = P/P_0$
- Works for monochromatic light
- Compound x has a unique e at different wavelengths

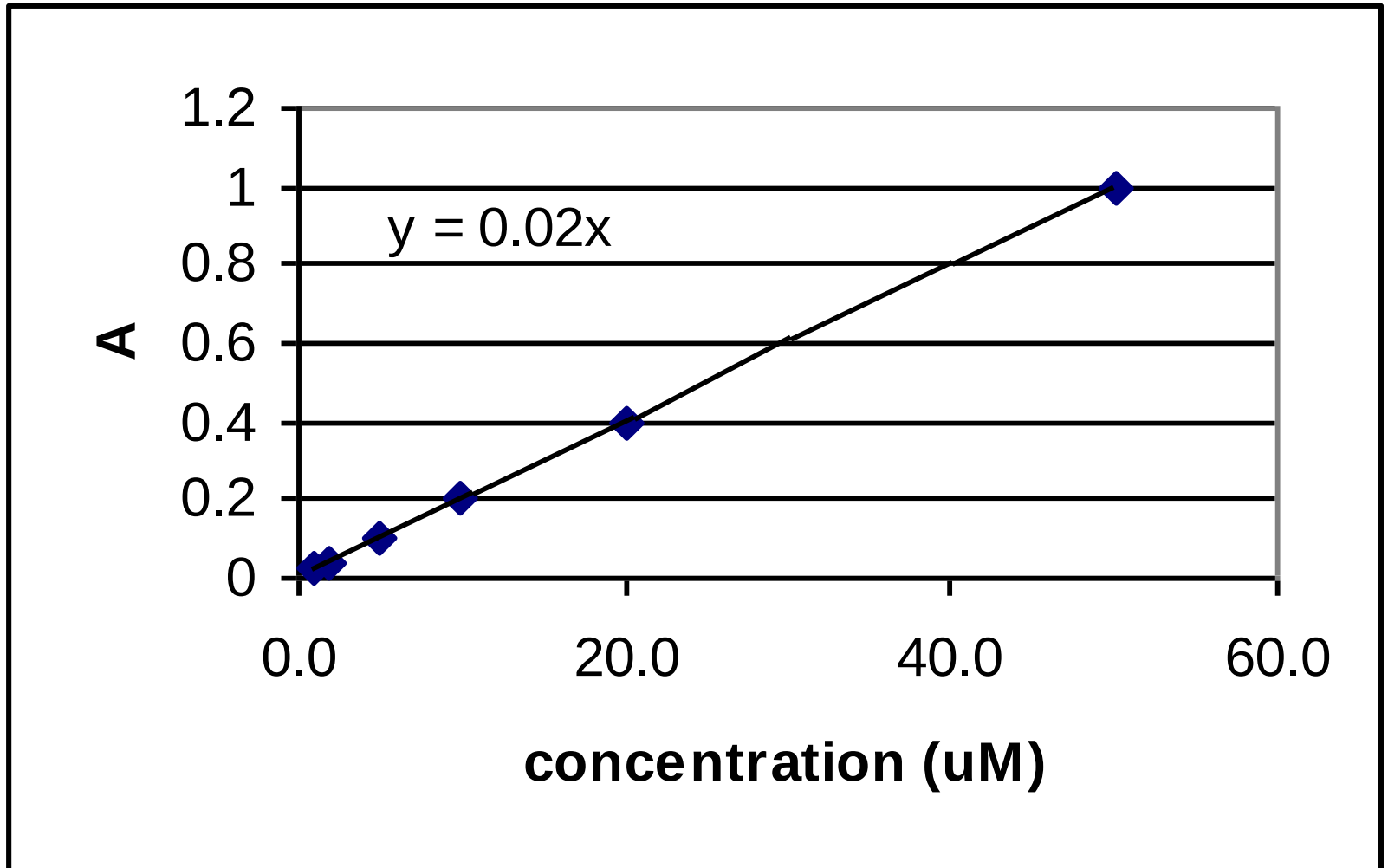
Characteristics of Beer's Law Plots

- One wavelength
- Good plots have a range of absorbances from 0.010 to 1.000
- Absorbances over 1.000 are not that valid and should be avoided
- 2 orders of magnitude

Standard Practice

- Prepare standards of known concentration
- Measure absorbance at λ_{max}
- Plot A vs. concentration
- Obtain slope
- Use slope (and intercept) to determine the concentration of the analyte in the unknown

Typical Beer's Law Plot

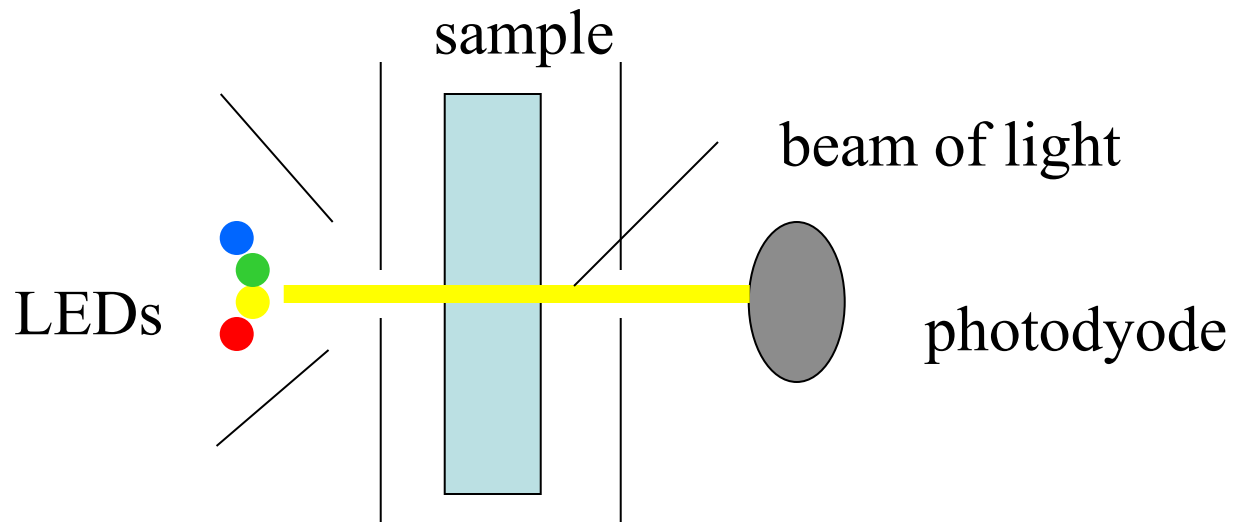


Instrumentation

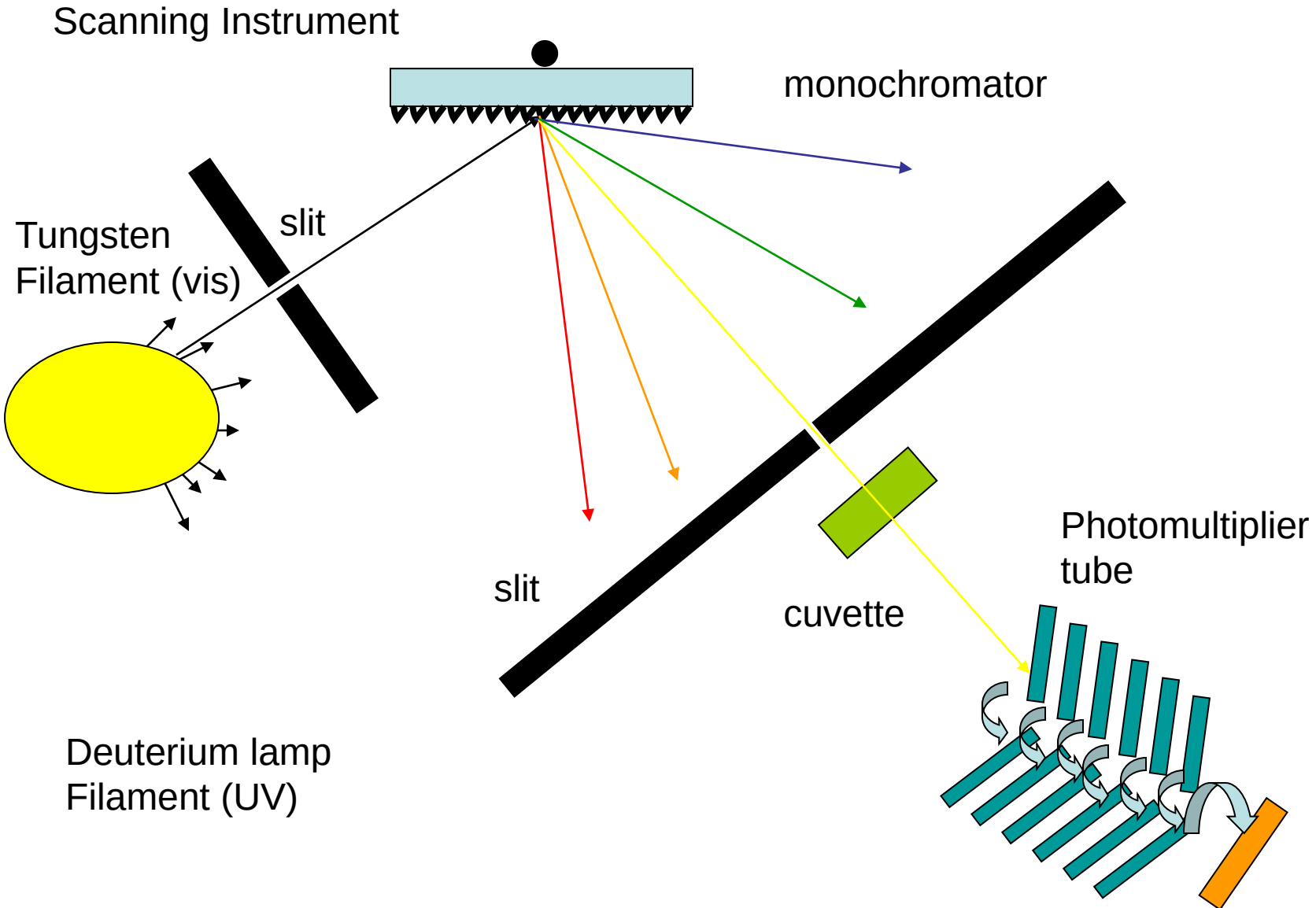
- Fixed wavelength instruments
- Scanning instruments
- Diode Array Instruments

Fixed Wavelength Instrument

- LED serve as source
- Pseudo-monochromatic light source
- No monochrometer necessary/ wavelength selection occurs by turning on the appropriate LED
- 4 LEDs to choose from



Scanning Instrument



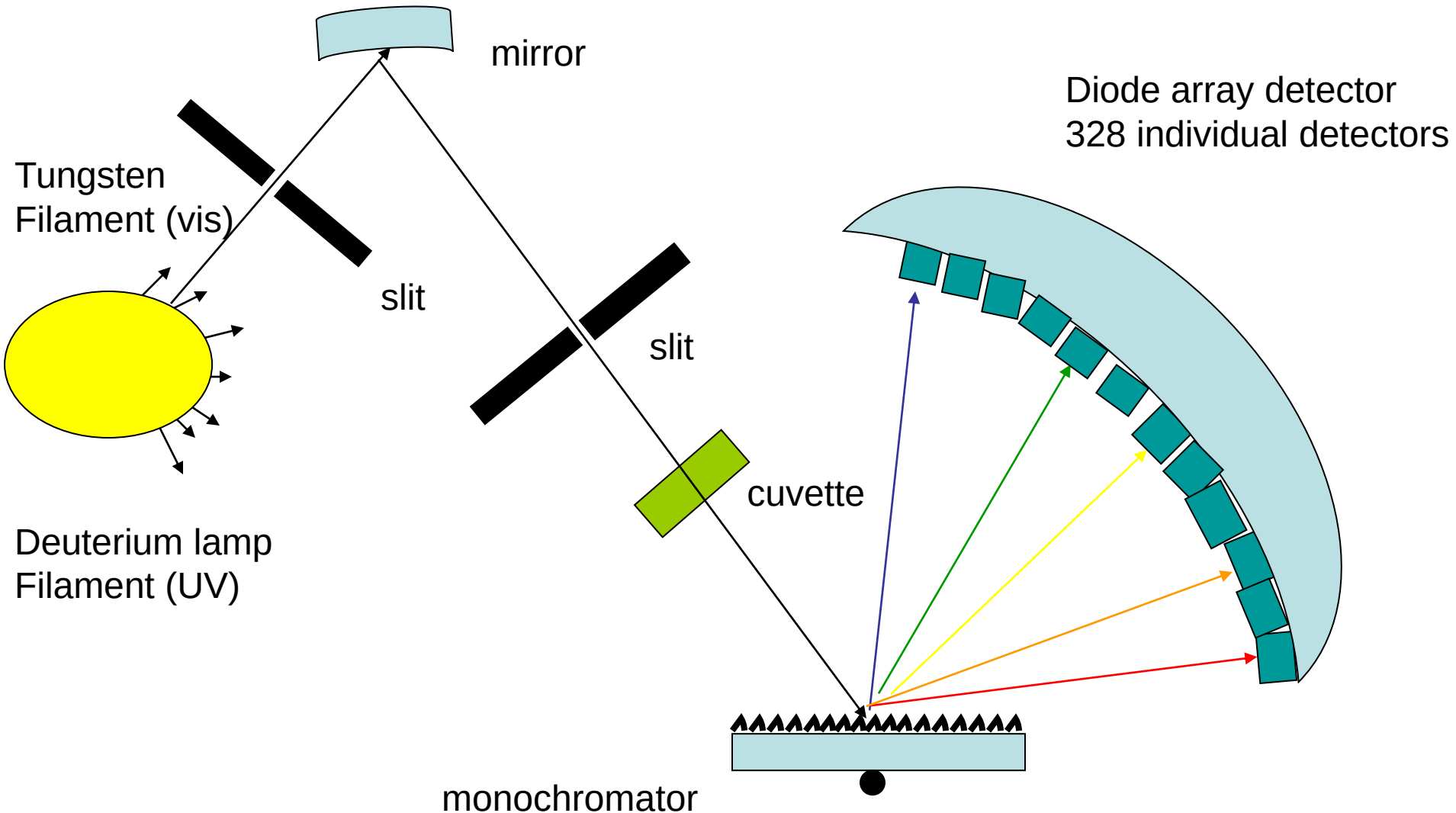
sources

- Tungsten lamp (350-2500 nm)
- Deuterium (200-400 nm)
- Xenon Arc lamps (200-1000 nm)

Monochromator

- Braggs law, $n\lambda = d(\sin i + \sin r)$
- Angular dispersion, $d\theta/d\lambda = n / d(\cos r)$
- Resolution, $R = \lambda/\Delta\lambda = nN$, resolution is extended by concave mirrors to refocus the divergent beam at the exit slit

Diode array Instrument



Advantages/disadvantages

- Scanning instrument
 - High spectral resolution (63000), $\lambda/\Delta\lambda$
 - Long data acquisition time (several minutes)
 - Low throughput
- Diode array
 - Fast acquisition time (a couple of seconds), compatible with on-line separations
 - High throughput (no slits)
 - Low resolution (2 nm)

HPLC-UV

